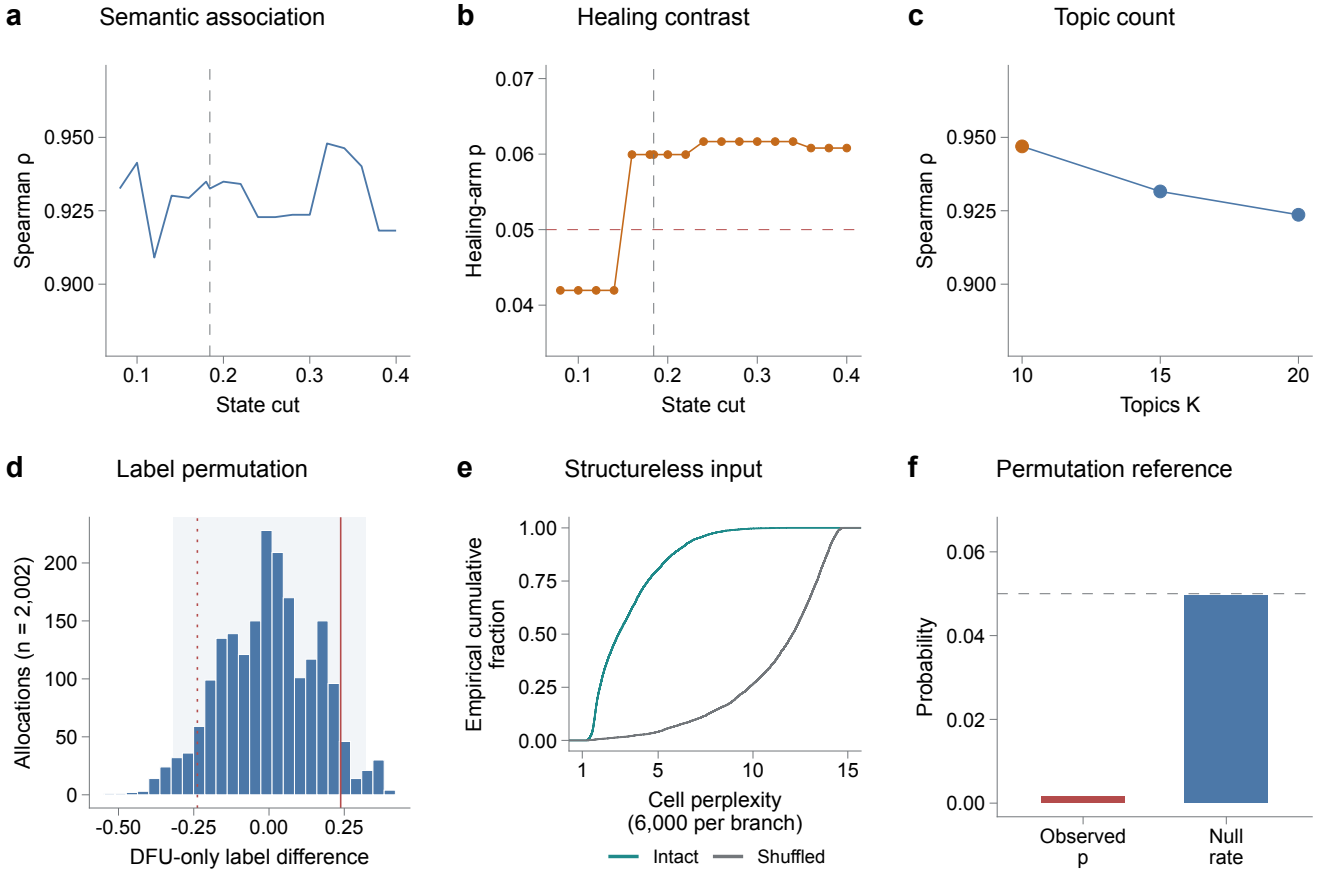

a
Semantic association
Spearman ρ
0.950
0.925
0.900
0.1
0.2
0.3
0.4
State cut
b
Healing contrast
Healing-arm p
0.07
0.06
0.05
0.04
0.1
0.2
0.3
0.4
State cut
c
Topic count
Spearman ρ
0.950
0.925
0.900
10
15
20
Topics K
d
Label permutation
Allocations (n = 2,002)
200
150
100
50
0
-0.50
-0.25
0.00
0.25
DFU-only label difference
e
Structureless input
Empirical cumulative fraction
1.00
0.75
0.50
0.25
0.00
1
5
10
15
Cell perplexity (6,000 per branch)
Intact
Shuffled
f
Permutation reference
Probability
0.06
0.04
0.02
0.00
Observed p
Null rate